

TOTAL RESPONSE OF SERIES RLC CIRCUIT

□ **Steady state response:** is the final value of $v(t)$

$$v_{ss}(t) = v(\infty) = V_s$$

The final value of the capacitor voltage is the same as the source voltage V_s (capacitor becomes open circuit in steady state)

□ Thus, the complete solution is,

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(Overdamped)}$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad \text{(Critically damped)}$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad \text{(Underdamped)}$$

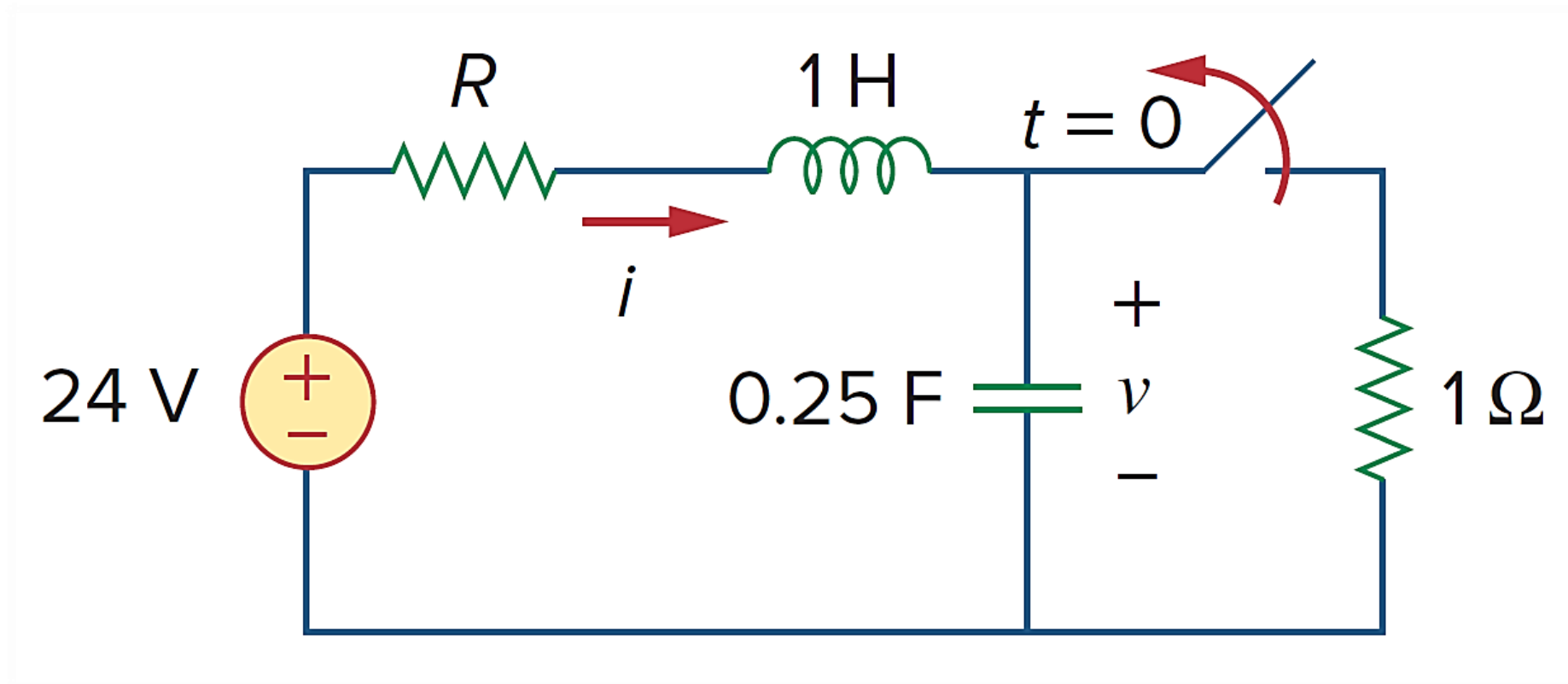
The values of the constants A_1 and A_2 are obtained from the initial conditions: $v(0)$ and $dv(0)/dt$.

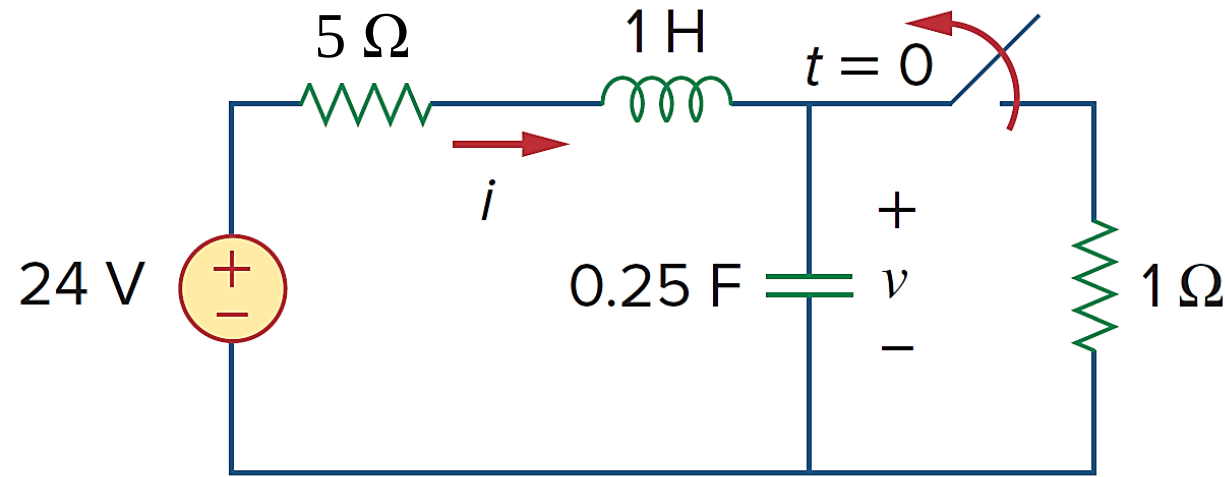
Numerical 1:

For the circuit shown, find $v(t)$ and $i(t)$ for $t > 0$.

Assume that the switch was closed for a long time before opening.

Consider three cases: (a) $R = 5\ \Omega$, (b) $R = 4\ \Omega$, and (c) $R = 1\ \Omega$.





Solution:

(a): $R = 5\ \Omega$

For $t < 0$, the capacitor behaves like an open circuit while the inductor acts like a short circuit. The initial current through the inductor is,

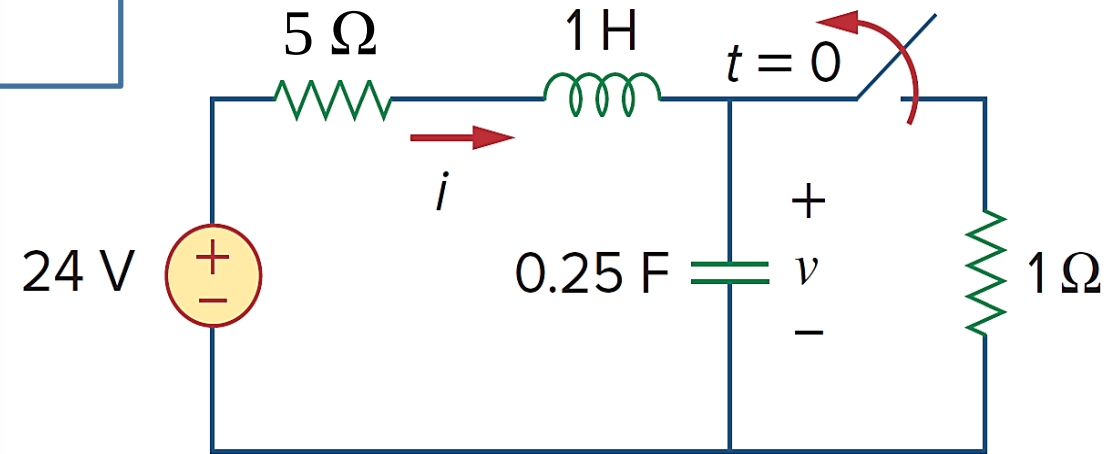
$$i(0) = \frac{24}{5+1} = 4A$$

and the initial voltage across the capacitor is the same as the voltage across the 1Ω resistor, $v(0) = 1i(0) = 4V$

For $t > 0$, the switch is opened, so that 1Ω resistor disconnected.
The characteristic roots are determined as follows:

$$\alpha = \frac{R}{2L} = \frac{5}{2 \times 1} = 2.5, \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{1 \times 0.25}} = 2$$

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -1, -4$$



Since $\alpha > \omega_0$, we have the **overdamped** natural response.

The total response is therefore $v(t) = v_{ss} + (A_1 e^{-t} + A_2 e^{-4t})$

v_{ss} is the final value of the capacitor voltage = **24 V**.

$$\Rightarrow v(t) = 24 + (A_1 e^{-t} + A_2 e^{-4t})$$

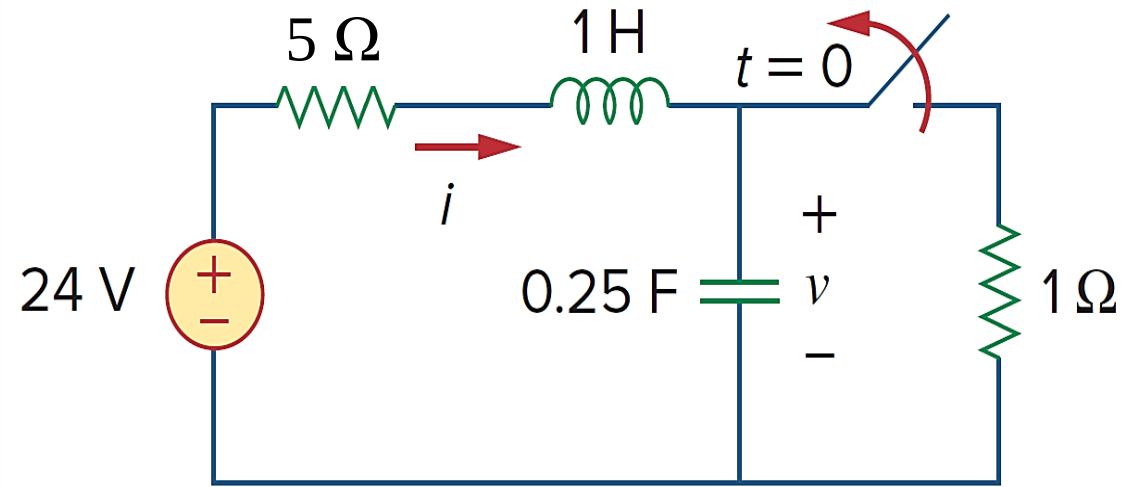
Determine A_1 and A_2 via initial conditions:

$$v(0) = 4 = 24 + A_1 + A_2 \Rightarrow -20 = A_1 + A_2$$

The current through the inductor cannot change abruptly and is the same current through the capacitor at $t = 0^+$ because the inductor and capacitor are now in series. Hence,

$$i(0) = C \frac{dV(0)}{dt} = 4, \frac{dV(0)}{dt} = \frac{4}{C} = \frac{4}{0.25} = 16$$

$$\frac{dV(0)}{dt} = 16 = -A_1 - 4A_2 \Rightarrow A_1 = -64/3, A_2 = 4/3$$



$$v(t) = 24 + \frac{4}{3}(-16e^{-t} + e^{-4t})$$

$$i(t) = C \frac{dV}{dt}$$

$$i(t) = \frac{4}{3}(4e^{-t} - e^{-4t})$$

CASE 2: $R = 4 \Omega$

The initial current through the inductor is $i(0) = \frac{24}{4+1} = 4.8A$
and the initial voltage across the capacitor is $v(0) = 1i(0) = 4.8V$

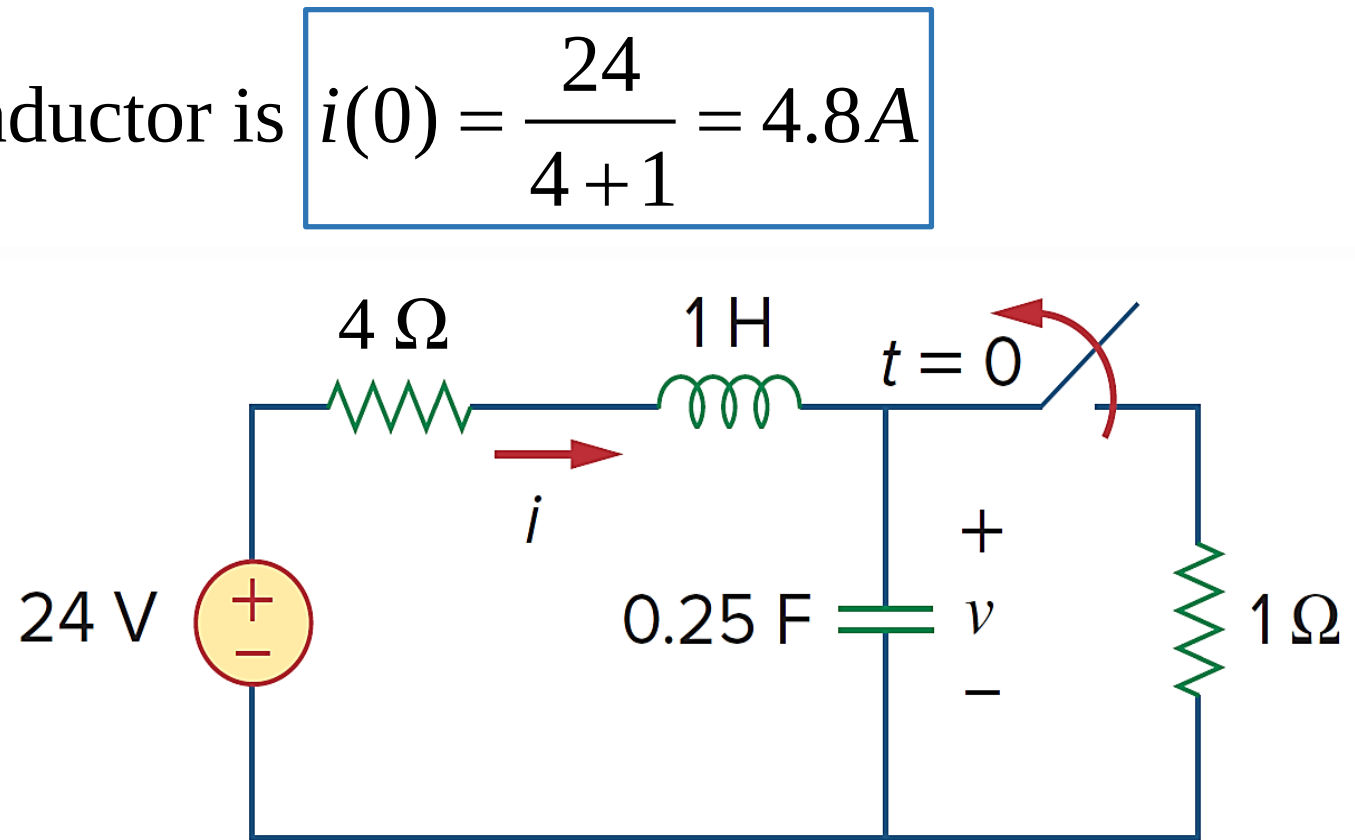
For the characteristic roots,

$$\alpha = \frac{R}{2L} = \frac{4}{2 \times 1} = 2$$

$$\omega_0 = 2$$

$s_1 = s_2 = -\alpha = -2 \Rightarrow$ **critically damped** response

$$v(t) = v_{ss} + (A_1 + A_2 t)e^{-2t} \Rightarrow v(t) = 24 + (A_1 + A_2 t)e^{-2t}$$



To find A_1 and A_2 , we use the initial conditions

$$v(0) = 4.8 = 24 + A_1 \Rightarrow A_1 = -19.2$$

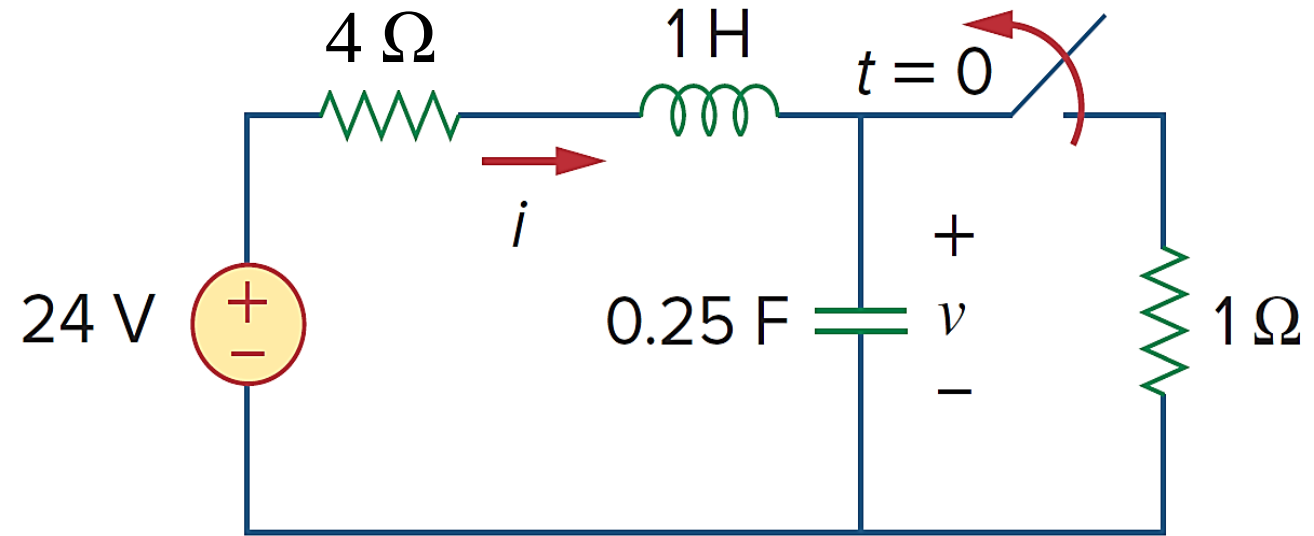
Since $i(0) = C dv(0)/dt = 4.8$,

$$\frac{dV(0)}{dt} = \frac{4.8}{C} = 19.2$$

$$\frac{dV}{dt} = (-2A_1 - 2tA_2 + A_2)e^{-2t}$$

$$\frac{dV(0)}{dt} = 19.2 = -2A_1 + A_2 \Rightarrow A_2 = -19.2$$

Thus, $v(t) = 24 - 19.2(1 + t)e^{-2t}$



The inductor current is the same as the capacitor current $i(t) = Cdv(t)/dt$
Multiplying by $C = 0.25$ and substituting the values of A_1 and A_2 into dv/dt

$$i(t) = (4.8 + 9.6t)e^{-2t}$$

CASE 3: $R = 1\ \Omega$

The initial inductor current is $i(0) = \frac{24}{1+1} = 12\text{ A}$

and the initial voltage across the capacitor is the same as the voltage across the $1\ \Omega$ resistor,

$$v(0) = 1i(0) = 12\text{ V}$$

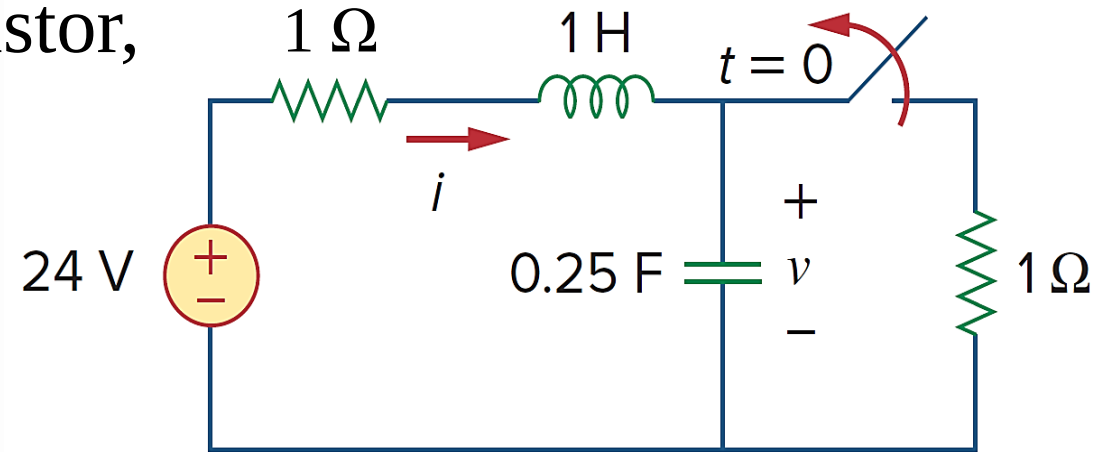
$$\alpha = \frac{R}{2L} = \frac{1}{2 \times 1} = 0.5$$

Since $\alpha = 0.5 < \omega_0 = 2$, we have the **under-damped** response

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -0.5 \pm j1.936$$

The total response is therefore,

$$v(t) = 24 + (A_1 \cos 1.936t + A_2 \sin 1.936t)e^{-0.5t}$$



In order to determine A_1 and A_2 ,

$$v(0) = 12 = 24 + A_1 \Rightarrow A_1 = -12$$

Since, $i(0) = C \, dv(0)/dt = 12$,

$$\frac{dv(0)}{dt} = \frac{12}{C} = 48$$

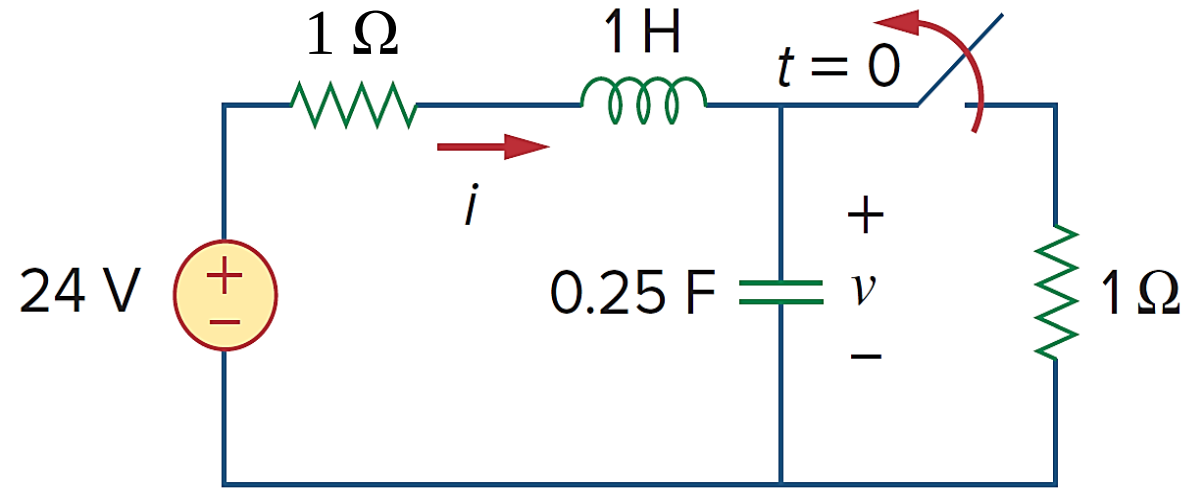
$$\frac{dv}{dt} = e^{-0.5t}(-1.936A_1 \sin 1.936t + 1.936A_2 \cos 1.936t) - 0.5e^{-0.5t}(A_1 \cos 1.936t + A_2 \sin 1.936t)$$

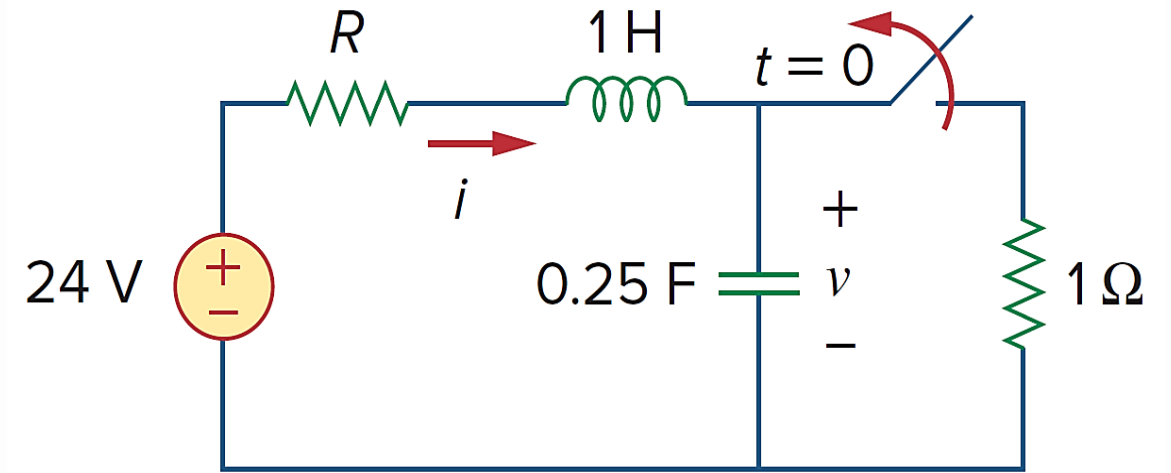
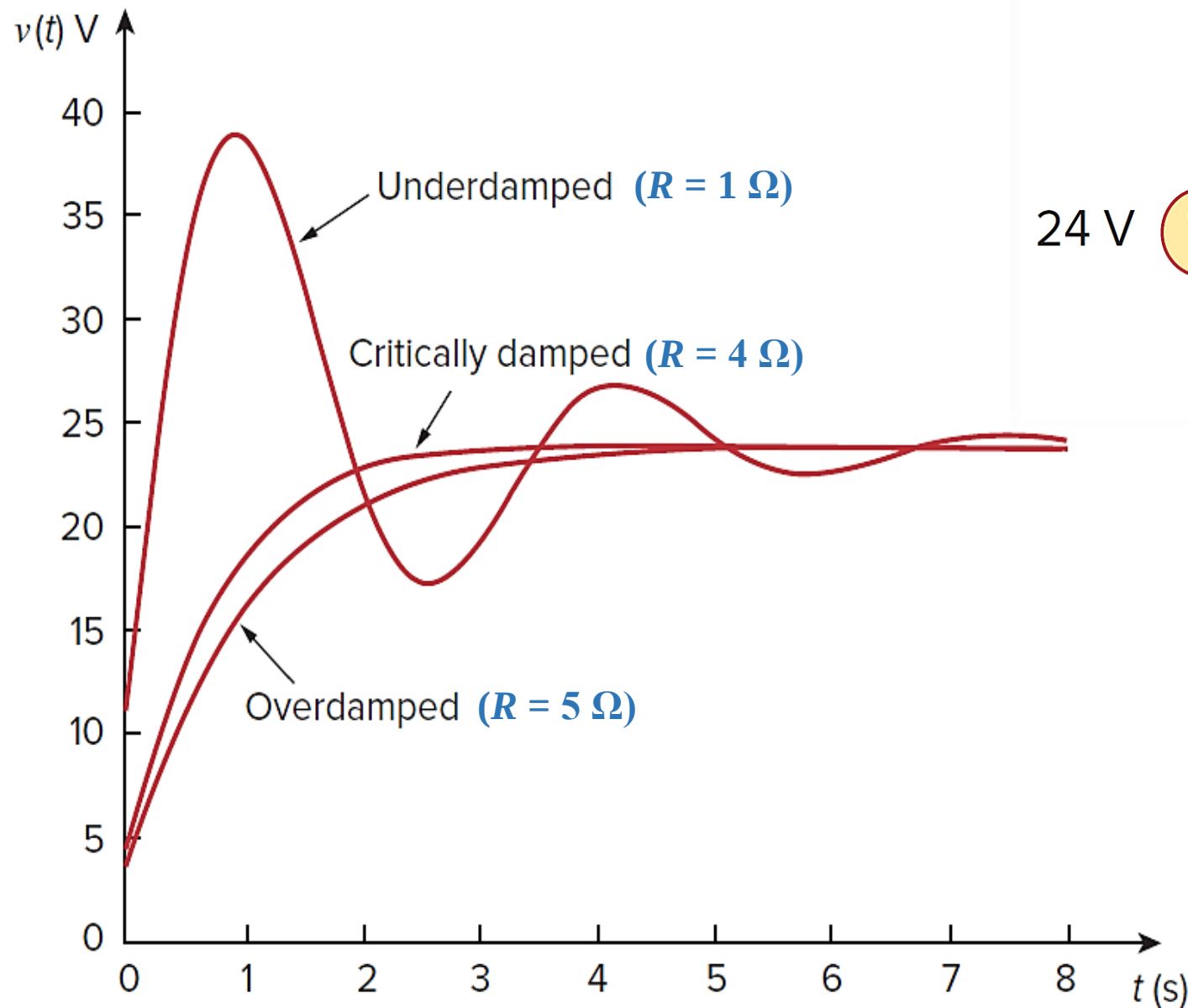
$$\frac{dv(0)}{dt} = 48 = (-0 + 1.936A_2) - 0.5(A_1 + 0) \quad \text{Substituting } A_1 = -12 \text{ gives } A_2 = 21.694$$

$$v(t) = 24 + (21.694 \sin 1.936t - 12 \cos 1.936t)e^{-0.5t}$$

Multiplying by $C = 0.25$ and substituting the values of A_1 and A_2 into dv/dt

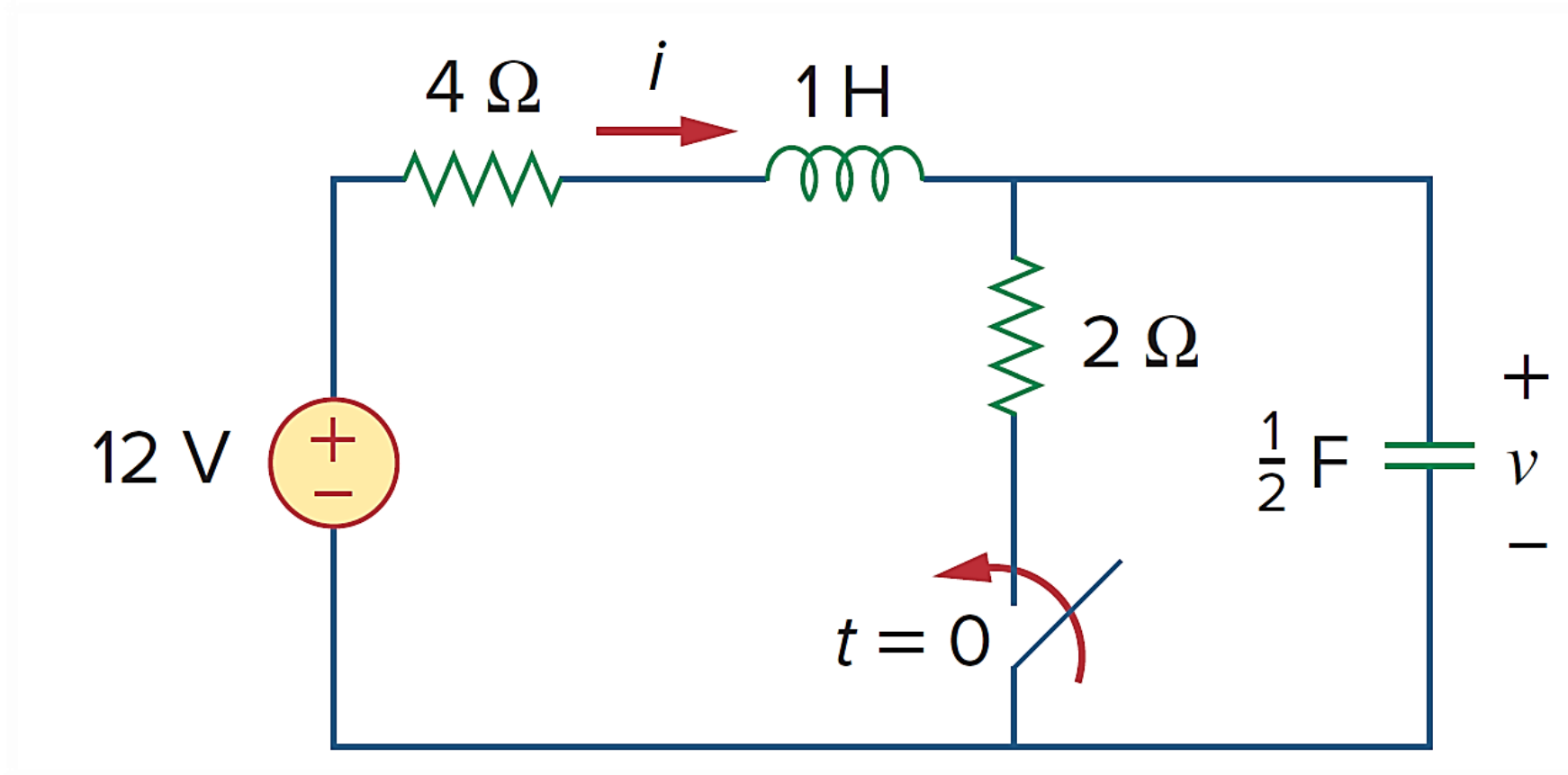
$$i(t) = (3.1 \sin 1.936t + 12 \cos 1.936t)e^{-0.5t}$$





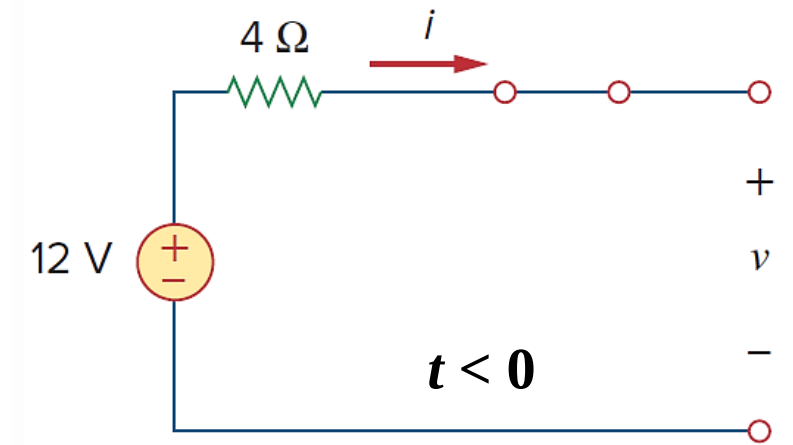
Critically damped circuit
gives fastest response
without any over-shooting

Numerical 3: Find the complete response $v(t)$ and $i(t)$ for $t > 0$. Assume that the switch is open for a long time before closing.



Solution: We first find the initial and final values,

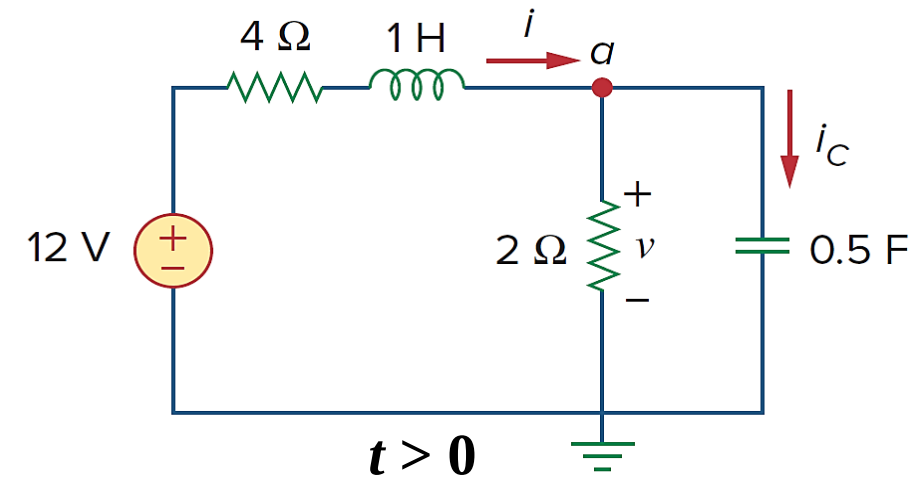
At $t = 0^-$, as shown the circuit is at steady state. $v(0^-) = 12V, i(0^-) = 0$



At $t = 0^+$, the switch is closed. By the continuity of capacitor voltage and inductor current,

$$v(0^+) = v(0^-) = 12V, i(0^+) = i(0^-) = 0$$

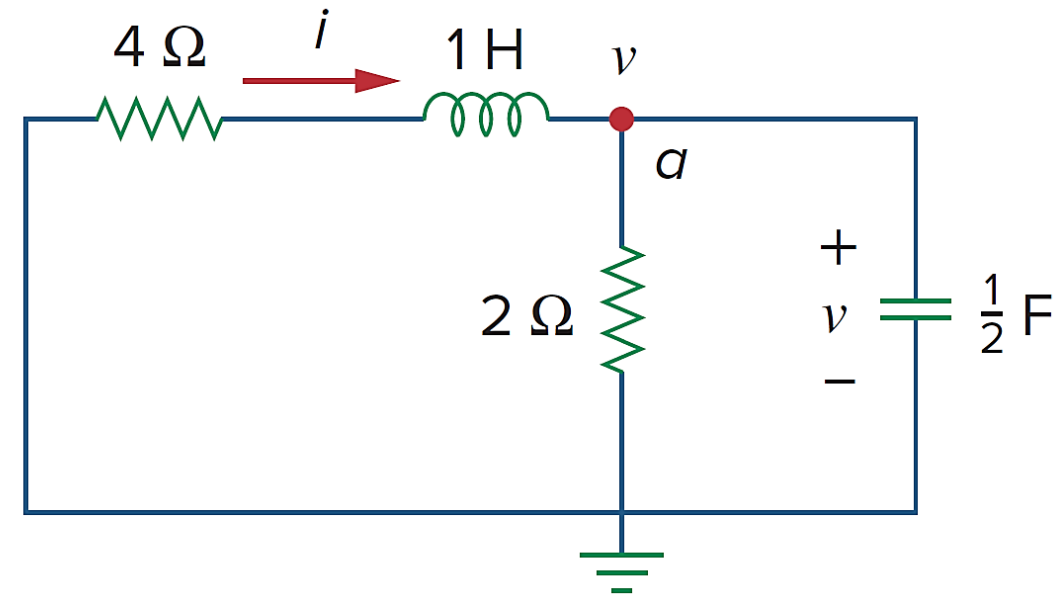
In order to determine $dv(0^+)/dt$, we use $dv/dt = i_C/C$. Applying KCL at node a ,



$$i(0^+) = i_c(0^+) + \frac{v(0^+)}{2} \Rightarrow 0 = i_c(0^+) + \frac{12}{2} \Rightarrow i_c(0^+) = -6A, \Rightarrow \frac{dv(0^+)}{dt} = \frac{-6}{0.5} = -12V/s$$

The final values are obtained when the inductor is replaced by a short circuit and the capacitor by an open circuit

$$i(\infty) = \frac{12}{4+2} = 2A, v(\infty) = 2i(\infty) = 4V$$



To get the transient response, we need to find the 2nd-order differential equation & corresponding characteristic equation

Applying KCL at node a gives,

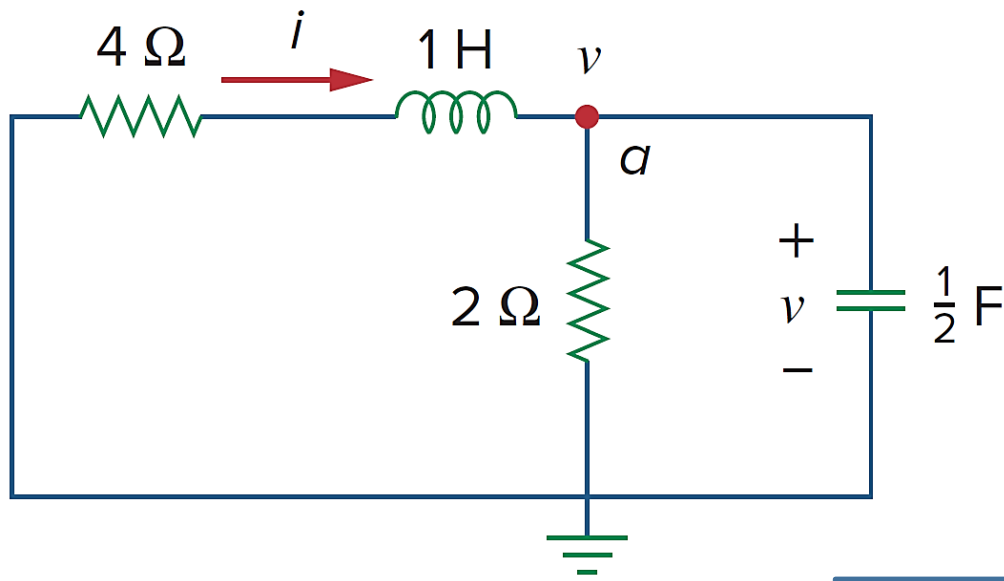
$$i = \frac{v}{2} + \frac{1}{2} \frac{dV}{dt},$$

Applying KVL in outer loop,

$$4i + 1 \frac{di}{dt} + v = 0$$

$$2v + 2 \frac{dv}{dt} + \frac{1}{2} \frac{dv}{dt} + \frac{1}{2} \frac{d^2v}{dt^2} + v = 0$$

$$\frac{d^2v}{dt^2} + 5 \frac{dv}{dt} + 6v = 0$$



The characteristic equation is obtained as,

$$s^2 + 5s + 6 = 0$$

with roots $s = -2$ and $s = -3$.

Thus, the natural response is

$$v_n(t) = Ae^{-2t} + Be^{-3t}$$

The complete response is $v(t) = v_t + v_{ss} = 4 + Ae^{-2t} + Be^{-3t}$

Use the initial values to determine A & B : $v(0) = 12 = 4 + A + B \Rightarrow A + B = 8$

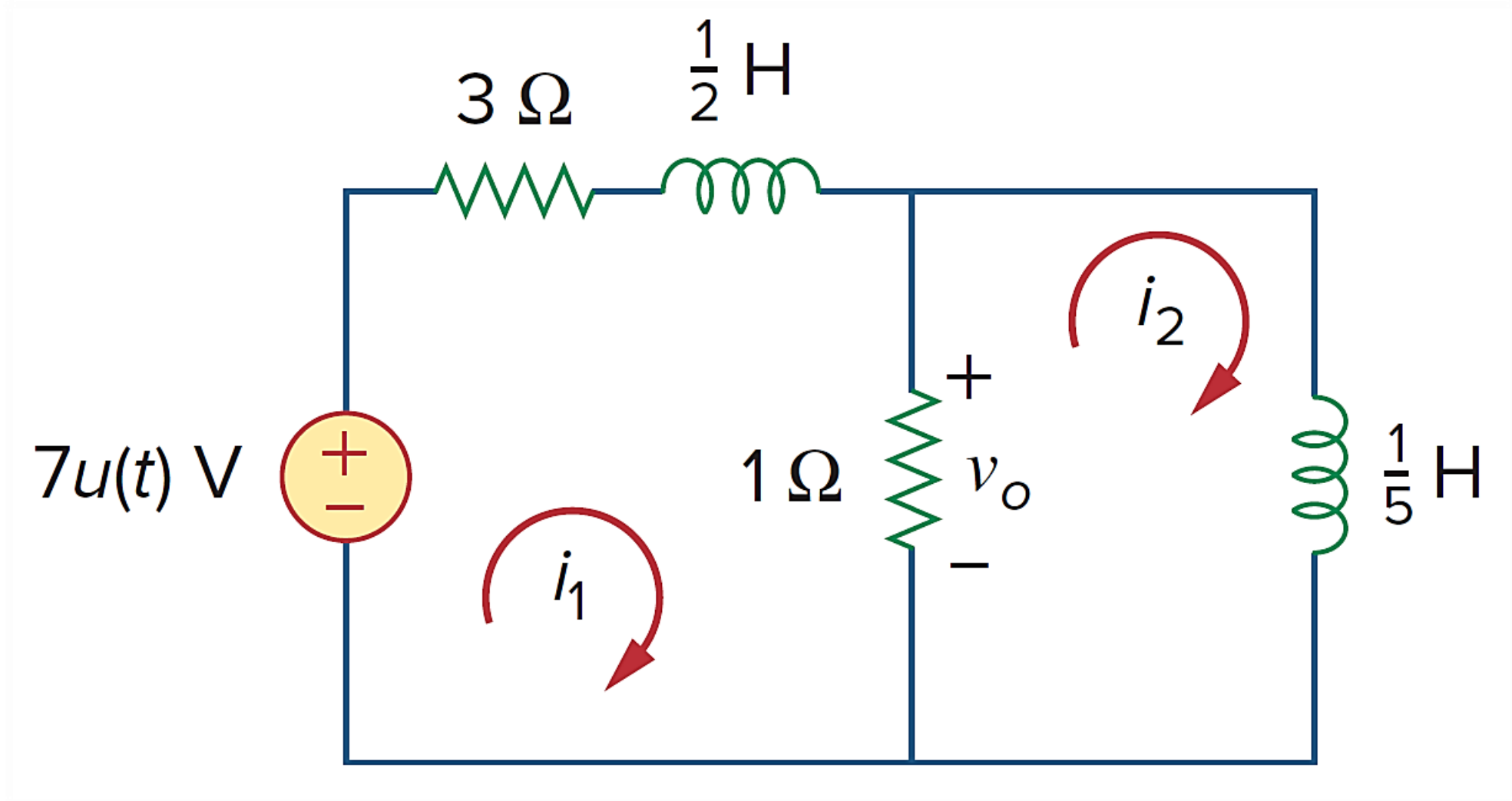
$$dv(0)/dt = -12 = -2A - 3B$$

Thus we obtain $A = 12, B = -4$

$$\Rightarrow v(t) = 4 + 12e^{-2t} - 4e^{-3t} \text{ V} \Rightarrow i = \frac{v}{2} + \frac{1}{2} \frac{dv}{dt} = 2 + 6e^{-2t} - 2e^{-3t} - 12e^{-2t} + 6e^{-3t}$$

$$= 2 - 6e^{-2t} + 4e^{-3t}$$

Numerical 4: Find $v_o(t)$ for $t > 0$ in the circuit shown



Solution:

For $t < 0$, $7u(t) = 0$, so that $i_1(0^-) = 0 = i_2(0^-)$. For $t > 0$, $7u(t) = 7$, so that the equivalent circuit is as shown,

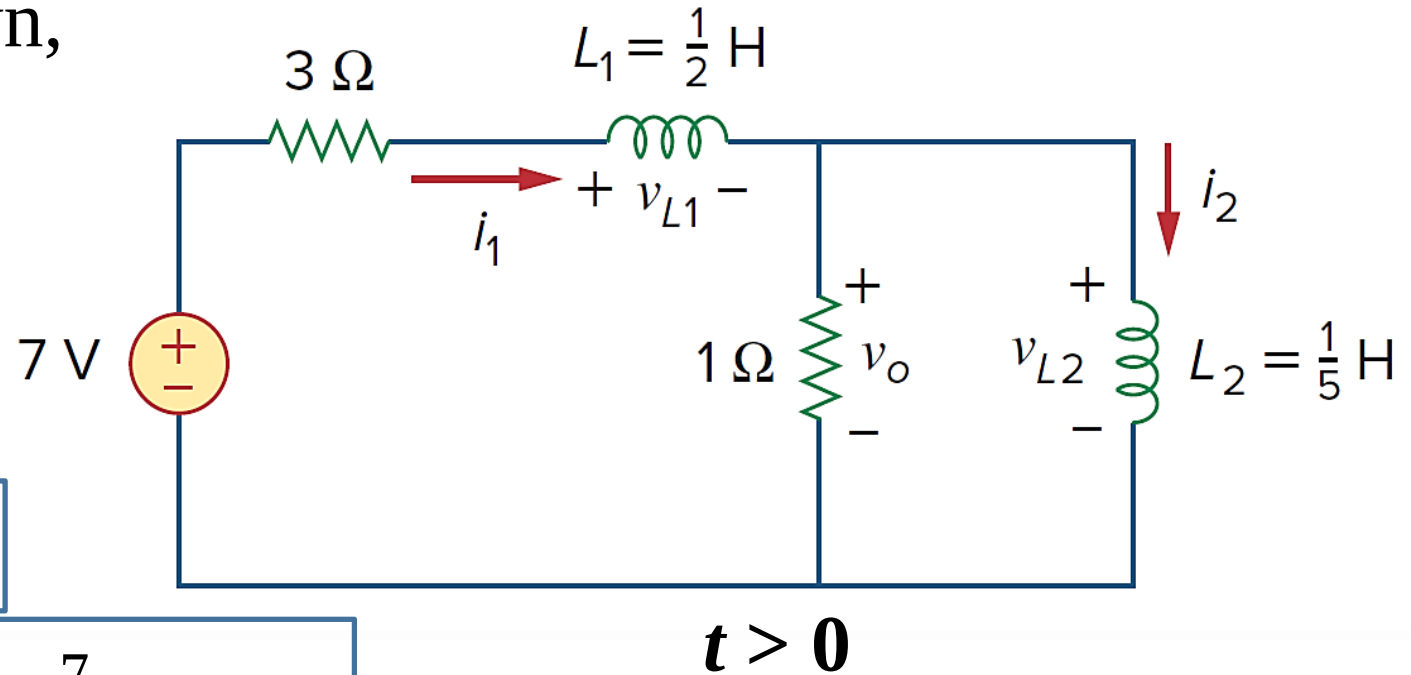
Due to the continuity of inductor current,

$$i_1(0^+) = i_1(0^-) = 0, i_2(0^+) = i_2(0^-) = 0$$

$$v_{L_2}(0^+) = v_o(0^+) = 1[i_1(0^+) - i_2(0^+)] = 0$$

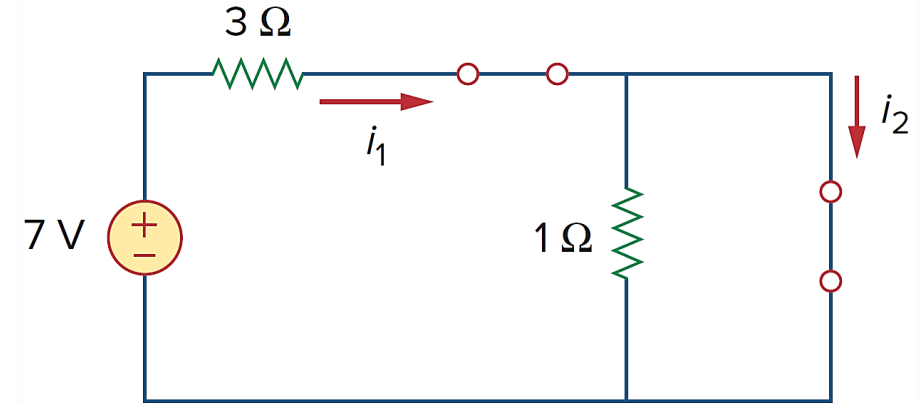
Since $L_1 \frac{di_1}{dt} = v_{L1}$, $\frac{di_1(0^+)}{dt} = \frac{v_{L1}}{L_1} = \frac{7}{\frac{1}{2}} = 14 \text{ A/s}$

Since $L_2 \frac{di_2}{dt} = v_{L2}$, $\frac{di_2(0^+)}{dt} = \frac{v_{L2}}{L_2} = 0$



As $t \rightarrow \infty$, the circuit reaches steady state, and the inductors can be replaced by short circuits as shown,

$$i_1(\infty) = i_2(\infty) = \frac{7}{3} \text{ A}$$



For transient response:

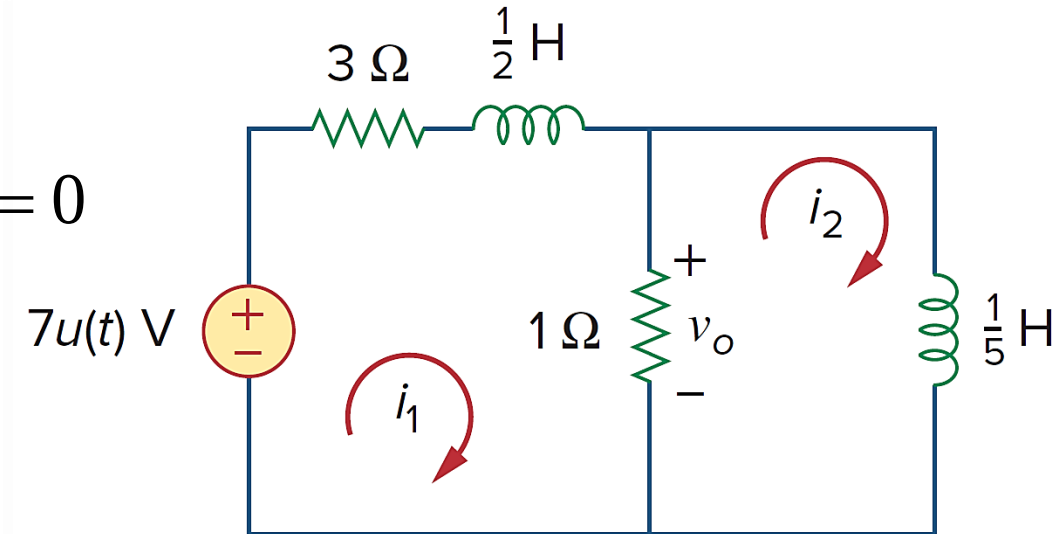
Apply KVL in the two meshes

$$4i_1 - i_2 + \frac{1}{2} \frac{di_1}{dt} = 0,$$

$$i_2 + \frac{1}{5} \frac{di_2}{dt} - i_1 = 0$$

$$i_2 = 4i_1 + \frac{1}{2} \frac{di_1}{dt} \Rightarrow 4i_1 + \frac{1}{2} \frac{di_1}{dt} + \frac{4}{5} \frac{di_1}{dt} + \frac{1}{10} \frac{d^2i_1}{dt^2} - i_1 = 0$$

$$\frac{d^2i_1}{dt^2} + 13 \frac{di_1}{dt} + 30i_1 = 0$$



The characteristic equation is determined as,

$$s^2 + 13s + 30 = 0$$

with roots $s = -3$ and $s = -10$

Hence, the transient response is,

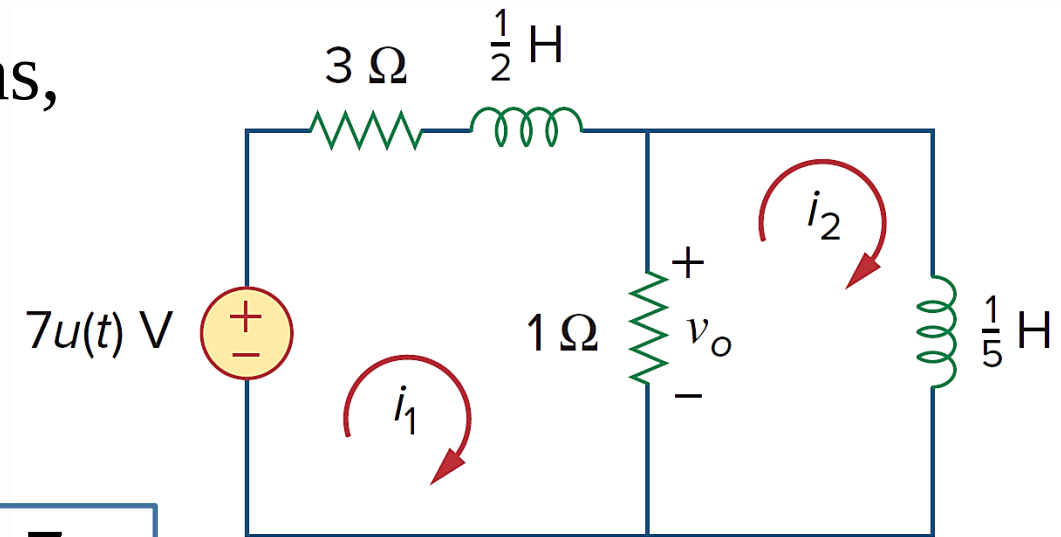
$$i_{1n} = Ae^{-3t} + Be^{-10t}$$

The steady-state response is, $i_{1ss} = i_1(\infty) = \frac{7}{3} A$

From initial conditions $i_1(0)$ and $di_1(0)/dt$: $0 = \frac{7}{3} + A + B, 14 = -3A - 10B$

$$\Rightarrow A = -4/3 \text{ and } B = -1$$

$$\text{Thus, } i_1(t) = \frac{7}{3} - \frac{4}{3}e^{-3t} - e^{-10t}$$



Applying KVL in left loop,

$$7 = 4i_1 - i_2 + \frac{1}{2} \frac{di_1}{dt}$$

$$\Rightarrow i_2 = -7 + 4i_1 + \frac{1}{2} \frac{di_1}{dt}$$

Substituting for i_1 gives,

$$i_2(t) = -7 + \frac{28}{3} - \frac{16}{3}e^{-3t} - 4e^{-10t} + 2e^{-3t} + 5e^{-10t} = \frac{7}{3} - \frac{10}{3}e^{-3t} + e^{-10t}$$

$$v_0(t) = 1[i_1(t) - i_2(t)], \Rightarrow v_0(t) = 2[e^{-3t} - e^{-10t}]$$

